Sign Frustration and Asset Dimension in DeFi Liquidation Clearing

Abstract. We study DeFi liquidation clearing with collateral-only sales and linear price impact. Even without debt-side buy pressure, clearing can be nonmonotone because selling an asset can lower another borrower's debt value and thereby rescue that borrower.

We give a worst-case structural complexity map controlled by the asset collateral-debt graph and the asset dimension. If the graph is bipartite and price impact is linear, a sign transform identifies smooth clearing with a bounded Z-matrix complementarity problem. Classical least-element complementarity theory then gives an exact polynomial-time clearing algorithm for arbitrary drift, including the large-impact regime. For sign-balanced separable markets, explicit monotone piecewise-linear impact also has an exact polynomial-time expansion. A different tractable regime appears when the number of assets is fixed. Then arbitrary linear instances are exactly solvable by enumerating the cells of a price-space hyperplane arrangement, and arbitrary constant-product instances admit a polynomial-time δ-clearing algorithm via fixed-dimensional real algebraic geometry. The same sign transform gives membership in CLS ($=$ PPAD $\cap$ PLS) for separable monotone nonlinear impact curves such as independent AMM pools, where the affine complementarity reduction no longer applies. In the other direction, non-bipartite instances can implement NOT gates and encode SUM/NOT generalized circuits, making constant-approximate clearing PPAD-hard. The hardness persists under independent constant-product AMM pricing. Smooth approximate clearing is in PPAD.

Keywords: DeFi liquidation · Clearing · PPAD · CLS · Tarski fixed points · Financial networks

1 Introduction and Technical Overview

DeFi lending protocols liquidate undercollateralized positions by repaying debt and selling collateral. When positions are large relative to liquidity, these collateral sales move prices, and the prices determine which positions are unhealthy. We isolate this collateral-sale channel in a conservative baseline model with external debt repayment and no debt-side buy pressure, because it is already sufficient to create the negative feedback and hardness below. Real liquidations can also create buy pressure in the debt asset. The sign-balanced tractability side is robust to a same-asset linear buy-pressure extension, while the hardness construction uses only the collateral-sale channel. The role-reversal behind our feedback is common, since major collateral assets and stablecoins can also be borrowed. Empirical DeFi studies document liquidation cascades and shared-asset fragility [15–18, 4].

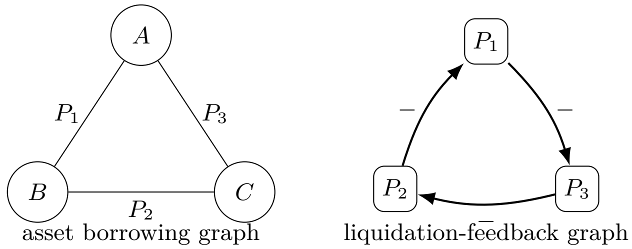

Fig. 1. A three-asset collateral-debt cycle. Selling an asset can lower another position's debt value, creating a negative liquidation dependency.

As a lightweight protocol calibration, we queried the Morpho public API on June 18, 2026 for Ethereum markets with the largest borrow balances [36]. In the resulting 700-market snapshot, the active asset borrowing graph is non-bipartite even after requiring at least $10,000 of both borrow and collateral exposure on each edge. One witness is LBTC – USDC – WBTC – LBTC. Appendix A gives the snapshot details and a Uniswap v3 stress calibration. On a mainstream USDC-WBTC-WETH odd cycle, selling 1% of the WBTC collateral exposure on the WBTC/USDC edge is 4.25% of the active WBTC-side Uniswap v3 reserve, giving a first-order local marginal-price gain $2f/\eta = 8.5$ at smoothing width $\eta = 1\%$. This is a relevance witness, not an empirical hardness claim.

A DeFi-native negative feedback. Consider three assets A, B, C and three positions

$$P_1 = (A, B), \qquad P_2 = (B, C), \qquad P_3 = (C, A),$$

where (A, B) denotes collateral A and debt B. Liquidating P_1 sells A, lowers p_A, and thereby lowers P_3's debt value, so liquidation can rescue another borrower. This all-negative feedback cycle, shown in Fig. 1, is the primitive behind both sign-balance and hardness.

Positioning. The novelty is not Tarski/lattice structure per se, nor the generic polynomial solvability of Z-matrix complementarity. The tractable linear side is a reduction. The DeFi sign transform identifies bipartite liquidation clearing with a classical bounded Z-matrix class. The new contribution is the collateral-debt sign structure, the asset-dimension phase diagram, and the collateral-only PPAD/AMM hardness boundary. Bipartiteness is a tractability guarantee, while non-bipartiteness is the worst-case structure used by our SUM/NOT hardness construction.

Main results. Let G_A be the asset borrowing graph, with assets as vertices and one edge per collateral-debt pair. If G_A is bipartite, flipping one side of prices and complementing the corresponding liquidation variables transforms the linear equations into

$$y = \Pi_{[0,u]}(Ay + c), \qquad A \geq 0.$$

For linear impact, this is a bounded Z-matrix complementarity problem, so classical least-element theory gives an exact polynomial-time algorithm for arbitrary drift and large impact. A second tractable regime is orthogonal. For fixed asset dimension n, arbitrary linear graphs are exactly solvable in $(m+n)^{O(n)}\,\mathrm{poly}(L)$ time by price-space arrangement enumeration, and arbitrary constant-product instances admit a $(m+n)^{O(n)}\,\mathrm{poly}(L, \log(1/\delta))$-time δ-algorithm. Explicit sign-balanced monotone PWL impact is also exact polynomial-time solvable, while general separable monotone impact lies in CLS. On the hard side, SUM/NOT gadgets preserve parity. NOT gates become odd collateral-debt paths, and SUM gates become even paths. Thus non-bipartiteness is intrinsic to the hard component. Constant-δ hardness holds for both linear and constant-product AMM pricing. These are worst-case statements. Small-impact or fixed-asset-dimension non-bipartite instances can still be easy.

2 Model and Sign Structure

There are assets $k \in [n]$, prices $p_k \in [0, M_k]$, and positions $i \in [m]$. Position i has collateral asset a_i, debt asset b_i, collateral amount $C_i > 0$, debt amount $D_i > 0$, liquidation threshold $\theta_i \geq 1$, and smoothing width $\varepsilon_i > 0$. We assume $a_i \neq b_i$. The health deficit is

$$h_i(p) = \theta_i D_i p_{b_i} - C_i p_{a_i}. \tag{1}$$

A positive deficit means the position is unhealthy. In the smooth model, liquidation intensity is

$$x_i = \Pi_{[0,1]}\left(\frac{h_i(p)}{\varepsilon_i}\right). \tag{2}$$

Liquidation sells collateral only. Thus the price equation is

$$p_k = \Pi_{[0,M_k]}\left(p_k^0 - s_k \sum_{i:a_i=k} \alpha_i x_i\right), \tag{3}$$

where $s_k > 0$ is the impact slope and $\alpha_i \geq 0$ is the collateral sold by a fully liquidated position. Fix a rational close factor $\varphi \in (0, 1]$, and require $0 \leq \alpha_i \leq \varphi C_i$. Debt is repaid externally and does not create a market buy order in the baseline model. Proposition 4 records a same-asset linear buy-pressure extension for the tractable sign-balanced side. The clipping makes the clearing map continuous on a compact box. The hardness construction respects the close factor by adjusting market depths s_k, not by overselling collateral.

A smooth exact clearing is a pair (p, x) satisfying the price and liquidation equations. A δ-clearing satisfies

$$\max\{\|p - P(x)\|_\infty, \|x - X(p)\|_\infty\} \leq \delta,$$

where P and X denote the right-hand sides of (3) and the liquidation equation.

All inputs are rational numbers encoded in binary. The search problem asks for a δ-approximate clearing. In the hardness theorem, δ is a fixed positive constant inherited from the source generalized-circuit gap and constant gadget amplification.

The asset borrowing graph G_A has assets as vertices and an undirected edge $\{a_i, b_i\}$ for each position. This is the graph used in the main statements.

Economic meaning of bipartiteness. Bipartiteness separates assets into two sectors, with every borrowing position crossing sectors, and rules out odd collateral/debt role-reversal cycles such as Fig. 1. With a bipartition, one global sign convention orders one side of prices normally and the other in reverse, giving the monotone system of Theorem 1. Non-bipartiteness is a worst-case source of hardness, not an instance-level iff criterion. Small-impact non-bipartite instances can still be contractions.

3 Sign-Balanced Clearing, Tarski, and Small-Impact Tractability

Theorem 1. *In the collateral-only linear price-impact model with price clipping, if the asset borrowing graph G_A is bipartite, then for every rational $\delta > 0$, smooth δ-clearing reduces in polynomial time to the succinct Tarski fixed-point problem. Hence bipartite smooth δ-clearing belongs to* $\mathrm{CLS} = \mathrm{PPAD} \cap \mathrm{PLS}$.

Let (L, R) be a bipartition of G_A. Define transformed price variables

$$q_k = \begin{cases} p_k, & k \in L, \\ M_k - p_k, & k \in R, \end{cases} \tag{4}$$

and transformed liquidation variables

$$z_i = \begin{cases} 1 - x_i, & a_i \in L, \\ x_i, & a_i \in R. \end{cases} \tag{5}$$

Let $y = (q, z)$ and let $[0, u]$ be the corresponding box, with upper bounds M_k for transformed prices and 1 for transformed liquidations.

Lemma 1. *Under the above transform, the clearing equations are equivalent to*

$$y = \Pi_{[0,u]}(Ay + c) \tag{6}$$

for a rational matrix $A \geq 0$ and vector c. Moreover, after ordering price and liquidation variables separately, A has block form

$$A = \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix}, \qquad P, Q \geq 0.$$

Proof. There are four cases. First, if $k \in L$, then every position with collateral k has $x_i = 1 - z_i$, and

$$q_k = p_k = \Pi_{[0,M_k]}\left(p_k^0 - \sum_{i:a_i=k} s_k\alpha_i + \sum_{i:a_i=k} s_k\alpha_i z_i\right),$$

which has nonnegative coefficients on the z_i's. Second, if $k \in R$, then $q_k = M_k - p_k$ and $x_i = z_i$ for positions collateralized by k. Using $M - \Pi_{[0,M]}(r) = \Pi_{[0,M]}(M - r)$,

$$q_k = \Pi_{[0,M_k]}\left(M_k - p_k^0 + \sum_{i:a_i=k} s_k\alpha_i z_i\right),$$

again with nonnegative coefficients.

For liquidations, consider $a_i \in L, b_i \in R$. Then $p_{a_i} = q_{a_i}$, $p_{b_i} = M_{b_i} - q_{b_i}$, and $z_i = 1 - x_i$. Since $1 - \Pi_{[0,1]}(r) = \Pi_{[0,1]}(1 - r)$,

$$z_i = \Pi_{[0,1]}\left(1 - \frac{\theta_i D_i M_{b_i}}{\varepsilon_i} + \frac{\theta_i D_i}{\varepsilon_i} q_{b_i} + \frac{C_i}{\varepsilon_i} q_{a_i}\right).$$

Finally, if $a_i \in R, b_i \in L$, then $z_i = x_i$, $p_{a_i} = M_{a_i} - q_{a_i}$, and $p_{b_i} = q_{b_i}$, giving

$$z_i = \Pi_{[0,1]}\left(-\frac{C_i M_{a_i}}{\varepsilon_i} + \frac{\theta_i D_i}{\varepsilon_i} q_{b_i} + \frac{C_i}{\varepsilon_i} q_{a_i}\right).$$

All displayed coefficients of variables are nonnegative. Price equations depend only on transformed liquidation variables, and liquidation equations depend only on transformed price variables, which gives the block-off-diagonal form.

The same sign check also handles a separable same-asset linear debt-buy pressure term on the sign-balanced tractable side. See Proposition 4 in Appendix C.

The map

$$T(y) = \Pi_{[0,u]}(Ay + c)$$

is therefore isotone in the product order. This is the Tarski consequence of sign-balance. For linear impact it will be strengthened to an exact polynomial-time algorithm in Section 3.1. The CLS statement below records the robust monotone fixed-point upper bound that also survives the nonlinear extension in Appendix C.

Proof (Proof of Theorem 1). Choose a rational grid step $\eta \le \delta$ and define $T_\eta(y) = \lfloor T(y) \rfloor_\eta$ on the finite product grid $[0, u] \cap \eta\mathbb{Z}^d$. Isotonicity of T and coordinatewise floor makes T_η an order-preserving succinct grid map. Any fixed point satisfies $y \le T(y) < y + \eta$, hence is an η-approximate fixed point and transforms back to a δ-clearing. Thus the problem reduces to succinct Tarski, which lies in PPAD and PLS [9]. Since CLS = PPAD $\cap$ PLS [11], it lies in CLS.

The same proof covers bipartite separable monotone price-impact curves, including independent asset–numeraire constant-product pools. Appendix C gives the exact statement and proof. Routed, cross-asset, or path-dependent AMM execution is outside this extension.

Proposition 1 (explicit monotone PWL impact). *In the sign-balanced separable model, suppose each g_k is a continuous nonincreasing piecewise-linear rational function, explicitly given by rational breakpoints and slopes, and the total number of pieces is polynomial in the input size. Then the least clearing in the transformed product order is computable exactly in polynomial time.*

The reason is that every nondecreasing PWL response in transformed coordinates has a subtraction-free clipped-affine expansion:

$$h(t) = h(0) + \sum_{j=1}^{r} s_j \, \Pi_{[0,\tau_j - \tau_{j-1}]}\big(t - \tau_{j-1}\big), \qquad s_j \geq 0.$$

Introducing one auxiliary variable for each hinge turns the transformed clearing equations into $Y = \Pi_{[0,U]}(A'Y + c')$ with $A' \geq 0$, so Proposition 2 applies. Appendix C gives the full proof. For sign-balanced genuinely curved or succinctly represented monotone impact with unbounded asset dimension, whether the CLS upper bound can be improved to P, or is tight via CLS-hardness, remains open.

The standard small-impact M-matrix/contraction regime gives uniqueness and fast iteration under an explicit inverse-polynomial gap. See Proposition 5 and Proposition 8 in Appendix C. The exact linear-impact algorithm below does not need a spectral gap.

3.1 Polynomial Time for Subtraction-Free Clearing

The sign-balanced transformation produces the abstract *subtraction-free box fixed-point* problem

$$z = \Pi_{[0,u]}(Bz + g), \qquad B \geq 0, \quad u > 0,$$

where g is the transformed drift. Let z^* denote the least fixed point and write

$$L^* = \{i : z_i^* = 0\}, \qquad I^* = \{i : 0 < z_i^* < u_i\}, \qquad U^* = \{i : z_i^* = u_i\}.$$

The fixed-point condition is the bounded complementarity system

$$\begin{gathered} 0 \leq z \leq u, \qquad w = (I - B)z - g, \\ z_i = 0 \Rightarrow w_i \geq 0, \\ 0 < z_i < u_i \Rightarrow w_i = 0, \\ z_i = u_i \Rightarrow w_i \leq 0. \end{gathered}$$

Since $B \geq 0$, the matrix $I - B$ is a Z-matrix. Thus bipartite linear clearing lands in the classical least-element theory of bounded Z-matrix complementarity.

Relation to least-element complementarity theory. Bounded Z-matrix complementarity and least elements are classical. Relevant references include Cottle–Veinott [27], Mangasarian [28], Chandrasekaran [26], Pang's box case $0 \le x \le b$ [29], Ahn's upper-bounded iterations [30], Pang–Chandrasekaran pivoting [31], and the monograph of Cottle–Pang–Stone [33]. Thus polynomial-time solvability is not new as an algorithmic fact. Our tractable-side contribution is the reduction identifying bipartite collateral-only clearing with this class. To avoid relying on a particular standard-form LCP embedding, Proposition 2 uses the self-contained activation cascade of Appendix C.2. The citations explain the provenance of the method.

Proposition 2 (bipartite linear clearing as bounded Z-matrix LCP). *After the transform of Lemma 1, bipartite large-impact clearing with arbitrary drift is a bounded complementarity problem on $[0,u] \subset \mathbb{R}^d$ whose matrix $I - B$ is a Z-matrix. Its least solution exists and is computable in time polynomial in the input bit length, for arbitrary $\rho(B)$. Equivalently, the least fixed point of $z = \Pi_{[0,u]}(Bz+g)$ with $B \ge 0$ is polynomial-time computable. In the basic linear clearing model, $d = n + m$.*

Proof. The displayed system is exactly the upper-bounded Z-matrix complementarity problem for $I-B$ and $-g$. For a self-contained algorithm, use the activation cascade in Appendix C.2. The outer loop activates coordinates whose current drive is positive. Lemma 11 shows that at most d activations occur and that the returned point is the least fixed point. Each activated block has a strictly positive least fixed point, so Lemma 10 computes it using at most as many exact linear solves as the block size, together with exact M-matrix tests. The needed subcriticality fact is Lemma 7, and the exact rational M-matrix tests are summarized in Lemma 14. Hence the algorithm performs $O(d^2)$ rational linear solves/tests of dimension at most d. Gaussian elimination, linear feasibility for the M-matrix certificates, and the rational comparisons all have polynomial bit complexity. The fixed point of the monotone map is its least complementarity solution, giving the least clearing. This is the fixed-point specialization of the classical least-element methods of Chandrasekaran and Pang [26, 29].

Corollary 1 (exact polynomial-time clearing for bipartite linear markets). *In the collateral-only linear price-impact model with rational inputs, if the asset borrowing graph G_A is bipartite, then an exact smooth clearing can be computed in polynomial time.*

Proof. Apply Lemma 1, solve the transformed bounded Z-matrix problem by Proposition 2, and invert the sign transform.

Remark 1 (linear sign-balanced clearing and nonlinear impact). Corollary 1 supersedes the linear-impact CLS upper bound. The CLS route remains useful for separable nonlinear impact, where the affine complementarity reduction no longer applies.

Appendix C records the equivalent upper-bounded Z-matrix form and cascade details. This also explains why algorithms for conservation-like monotone payment networks [1] do not subsume the large-impact DeFi map: the transformed equations are not substochastic payment-allocation equations.

3.2 Fixed Asset Dimension

Sign-balance is not the only tractability resource. For linear price impact, bounded asset dimension gives an exact algorithm even on non-bipartite graphs, because the nonmonotone map is piecewise affine in price space.

Lemma 2 (price-space elimination). *In the linear smooth model, define*

$$x_i(p) = \Pi_{[0,1]}\big(h_i(p)/\varepsilon_i\big), \qquad h_i(p) = \theta_i D_i p_{b_i} - C_i p_{a_i},$$

$$F_k(p) = \Pi_{[0,M_k]}\Big(p_k^0 - s_k \sum_{i:a_i=k} \alpha_i\, x_i(p)\Big), \qquad F : \prod_k [0, M_k] \to \prod_k [0, M_k].$$

Then (p, x) is a clearing iff $p = F(p)$ and $x = x(p)$. This holds for an arbitrary asset graph G_A and requires no sign-balance assumption.

Proof. The liquidation equation makes each x_i a function of p_{a_i} and p_{b_i} alone. Substituting these functions into the price equation gives $p = F(p)$. Conversely, any fixed point p^* of F, together with $x(p^*)$, satisfies both the liquidation and price equations.

Theorem 2 (bounded asset dimension). *Let n be the number of assets and m the number of positions, with rational data of bit length L, on an arbitrary asset graph G_A. The clearing set can be represented as a union of $(m+n)^{O(n)}$ explicitly computable rational polyhedral pieces, each piece described by a rational linear system and inequalities. In time $(m+n)^{O(n)}\,\mathrm{poly}(L)$ one can enumerate all clearing-bearing pieces and output an exact rational clearing from each such piece. In particular, for each fixed n, linear smooth clearing is exact polynomial-time solvable on arbitrary graphs.*

Proof. By Lemma 2, fixed points live in n-dimensional price space. The $2m$ hyperplanes $h_i(p) = 0, \varepsilon_i$ partition this space into $(m+n)^{O(n)}$ faces, enumerable with rational sign descriptions in $(m+n)^{O(n)}\,\mathrm{poly}(L)$ time [10]. On a face, every x_i is one of $0, h_i(p)/\varepsilon_i, 1$, so the pre-clip price update is affine. Refining by the at most $2n$ price-clip hyperplanes makes the full map affine, $F(p) = Ap + b$, on each subface.

A clearing in such a subface is exactly a solution of $(I - A)p = b$ lying in the rational linear equalities and inequalities defining the subface. Exact Gaussian elimination plus rational linear feasibility decides this in $\mathrm{poly}(L)$ time and returns a rational point when feasible. Enumerating all $(m+n)^{O(n)}$ subfaces gives the claim. Appendix B gives the boundary and bit-complexity details.

Theorem 3 (bounded asset dimension with constant-product impact). *Replace the linear price equation by independent constant-product execution-price pools*

$$p_k = g_k(Q_k) = \min\left\{\frac{\kappa_k}{R_k + Q_k}, M_k\right\}, \qquad Q_k = \sum_{i:a_i=k} \alpha_i x_i,$$

with rational $R_k, \kappa_k, M_k > 0$, or by any bounded-degree rational impact curve, such as the constant-product marginal price $P_k(1+Q_k/R_k)^{-2}$, whose denominator is bounded away from zero on the relevant sales range. Let n be the number of assets and m the number of positions, with rational data of bit length L, on an arbitrary asset graph G_A. For every rational $\delta > 0$, a δ-clearing can be computed in time $(m+n)^{O(n)} \operatorname{poly}(L, \log(1/\delta))$. For each fixed n, this is polynomial in the input size and $\log(1/\delta)$.

Proof. The liquidation breakpoints are unchanged, so the same $(m + n)^{O(n)}$ price-space arrangement is used. On a face, each $Q_k(p)$ is rational affine. For a constant-product pool, the uncapped fixed-point equation is

$$p_k\big(R_k + Q_k(p)\big) = \kappa_k,$$

and the capped branch is $p_k = M_k$ plus the corresponding polynomial inequality. Thus each face and cap branch gives a bounded-degree semialgebraic system in the fixed number n of variables. Fixed-dimensional real-algebraic algorithms decide feasibility and return η-approximations in time polynomial in the coefficient size and $\log(1/\eta)$ [22, 2]. Since denominators are bounded below, F has an effective Lipschitz bound K_F. Choosing $\eta \le \delta/(1 + K_F)$ yields a δ-clearing. Appendix B gives the output-residual details.

Corollary 2 (two independent tractability resources). *Linear smooth clearing is polynomial-time solvable whenever either G_A is bipartite, for any number of assets, or the number of assets is fixed, for any graph. Constant-product smooth clearing is δ-solvable in polynomial time for fixed asset dimension, again on arbitrary graphs. Thus fixed asset dimension rules out the PPAD-hardness mechanism in both price models. In the linear model, bipartiteness is the other tractability guarantee. In the constant-product model, our hard family uses both sign frustration and growing asset dimension. Whether the unbounded sign-balanced nonlinear class is in* P *or is CLS-hard remains open.*

Remark 2. The bounds $(m+n)^{O(n)} \operatorname{poly}(L)$ and $(m+n)^{O(n)} \operatorname{poly}(L, \log(1/\delta))$ are fixed-dimension polynomials, not FPT bounds $f(n) \operatorname{poly}(m, L)$. On non-bipartite graphs F need not be monotone, so there need not be a least clearing. The arrangement algorithm returns clearing representatives cell by cell. In the bipartite linear case, Proposition 2 directly returns the least clearing; the arrangement method is an alternative exact solver for arbitrary graphs. Unlike Theorem 2, Theorem 3 outputs a δ-clearing rather than an exact rational clearing, because constant-product fixed points can be algebraic. The same proof extends to bounded-degree rational impact curves whose denominators are bounded away from zero on the relevant sales intervals.

4 Smooth Membership and PPAD-Hardness

Proposition 3. *For rational inputs and rational $\delta > 0$, smooth δ-clearing belongs to PPAD.*

Proof. For linear price impact, the right-hand sides of the price and liquidation equations are represented by a rational piecewise-linear circuit over addition, multiplication by rational constants, and min/max gates. The price clipping and liquidation clipping make this circuit a continuous self-map of a compact rational box. A δ-clearing is precisely an approximate fixed point of this piecewise-linear map, up to a harmless rescaling of residuals between price and liquidation coordinates. Approximate fixed points of rational piecewise-linear FIXP circuits are in PPAD via the Linear-FIXP/PPAD correspondence [14]. For constant-product curves, the map is instead a rational algebraic circuit whose denominators are bounded below by positive reserves on the compact domain, so the polynomial-continuity form of the same PPAD membership theorem applies [14, Proposition 2].

Our hardness proof uses only two generalized-circuit gates. Let $T_{[0,1]}$ denote clipping to $[0, 1]$. A κ-approximate assignment gives each wire a value in $[0, 1]$ and makes every gate output differ from its prescribed value by at most κ. This is a local notion of satisfaction. Circuits may contain cycles, and no topological evaluation order is assumed.

Lemma 3 (SUM/NOT source, Budish et al.). *There is a constant $\kappa > 0$ such that finding a κ-approximate solution to generalized circuits with only the gates*

$$G_+(u, w) : y_v = T_{[0,1]}(y_u + y_w), \qquad G_\neg(u) : y_v = 1 - y_u$$

is PPAD-complete, even with fan-out two.

Budish et al. [8, Appendix D.1, Definitions 10–11 and Theorem 3] prove this for the simplified generalized-circuit problem with fan-out two. Their definition allows wire values in $[0, 1 + \kappa]$. The restricted $[0, 1]$ formulation used here remains hard: after fixing unproduced source nodes in $[0, 1]$, the resulting clipped SUM/NOT map has an exact fixed point in $[0, 1]$, and every $[0, 1]$-valued κ-solution is also valid for their formulation. Related restricted-gate hardness appears in Filos-Ratsikas et al. [7]. Since truncated-SUM is already primitive, our gadgets need no high-gain chains, comparison gates, or band resets. Corollary 3 shows that the sign-balanced SUM/NOT fragment is in P. Hardness therefore has to come from odd-NOT sign imbalance.

Lemma 4 (unbalanced SUM/NOT source). *Lemma 3 remains PPAD-hard under the promise that the signed dependency graph of the source circuit contains a cycle with an odd number of NOT gates in the same connected component as an original hard-instance wire.*

Proof. Given a SUM/NOT circuit C, pick any wire s in it. Add two fresh wires a, b and the two gates

$$a = 1 - s, \qquad b = T_{[0,1]}(s + a).$$

Every solution of the augmented circuit restricts to a solution of C, because the original gates are unchanged. Conversely, any exact solution of C extends by setting $a = 1 - s$ and $b = 1$. The new dependency graph contains the cycle $s - a - b - s$, with the edge $s - a$ coming from a NOT gate and the two edges incident to b coming from the SUM gate. Thus the cycle has odd NOT parity and lies in the connected component containing s. A κ-approximate solution of the augmented circuit gives a κ-approximate solution of C by projection, so a solver for this promised subclass would solve the PPAD-hard source problem.

Wire encoding and writer equation. For each circuit wire v, create an asset and decode $\widehat{y}_v = \Pi_{[0,1]}(2 - p_v)$, with intended prices in $[1, 2] \subset [0, 3]$. In the exact gadget algebra below we write the raw value as $y_v = 2 - p_v$. A collateral-only writer with collateral output O, debt input U, parameters $C, D, \varepsilon, \theta = 1$, baseline $p_O^0 = 2$, and effective output weight $s_O\alpha = 1$ satisfies

$$x = \Pi_{[0,1]}(c - ay_U + \lambda y_O), \qquad a = D/\varepsilon, \quad \lambda = C/\varepsilon, \quad c = (2D - 2C)/\varepsilon. \quad (7)$$

If this writer is the unique effective writer of O, then $y_O = x$. In the linear sheet all gadgets have self coefficient at most $1/2$, so each local fixed point is unique and residuals are amplified only by a universal constant. Average co-writers use effective output weight $1/2$ each. The AMM sheet of Appendix E retunes the parameters after lifting the price baseline; its gain-2 self coefficient is $< 2/3$, still bounded away from one. Close-factor compliance is mechanical. A desired output weight η is implemented by taking $\alpha = \varphi/2$ and output slope $s_O = 2\eta/\varphi$. Appendix D records the fixed linear parameter sheet.

The two gadgets. The NOT gate $y_O = 1 - y_U$ uses one writer with $C = 1, D = 2, \varepsilon = 3$, giving

$$y_O = \Pi_{[0,1]}\left(\frac{2 - 2y_U + y_O}{3}\right),$$

whose unique fixed point is $1 - y_U$. The truncated-SUM gate is a constant-size macro. First, two co-writers, each contributing half a unit of price drop and reading the complement wires $\bar{U}, \bar{W}$, create an average asset A satisfying $y_A = (y_U + y_W)/2$. Second, a NOT gadget creates $\bar{A}$. Finally, one writer with $C = 1, D = 2, \varepsilon = 2$ reads $\bar{A}$ and satisfies

$$y_O = \Pi_{[0,1]}\left(y_A + \frac{1}{2}y_O\right),$$

whose unique fixed point is $T_{[0,1]}(2y_A) = T_{[0,1]}(y_U + y_W)$. The algebra and constant error bounds are recorded in Appendix D. In particular, a source gate violation is at most $B\delta$ for an absolute constant B independent of the circuit size.

Lemma 5 (forward decoding). *For every SUM/NOT circuit C, one can construct in polynomial time a collateral-only liquidation instance I_C such that every δ-clearing of I_C decodes via $\widehat{y}_v = \Pi_{[0,1]}(2 - p_v)$ to a $B\delta$-satisfying assignment of C, where one may take $B = 30$, independent of $|C|$.*

Proof. Replace each source gate by the corresponding constant-size macro. Debt-leg fan-out is non-invasive because only collateral sales enter (3). Every gate-output wire has a unique effective writer, except for the two intended co-writers in the average macro. Since each macro has constant size and self coefficient bounded away from one, the local decoding error is at most $B\delta$ for a universal B. Generalized-circuit satisfaction is local, so cycles require no topological induction.

The target search problem is total by Brouwer, because clipping makes the clearing map continuous on a compact box. A total-search reduction therefore needs only forward decoding.

Lemma 6 (gadget parity). *Let H_C be the signed undirected dependency graph of a SUM/NOT circuit C, with a negative edge for every NOT gate and positive edges from each input of a SUM gate to its output. For the liquidation instance I_C constructed above, with no additional disconnected anchor, the asset borrowing graph $G_A(I_C)$ is bipartite if and only if H_C is sign-balanced. In particular, an odd-NOT cycle in C maps to an odd cycle in $G_A(I_C)$.*

Proof. The claim is a parity accounting for the templates. A NOT gate is one collateral-debt edge, hence odd. A SUM input incidence is the even path

$$U - \bar{U} - A - \bar{A} - O,$$

and the same holds for $W \to O$. The remaining internal SUM subgraph is bipartite. Hence $G_A(I_C)$ is obtained from H_C by replacing negative edges by odd paths and positive edges by even paths, plus bipartite subdivisions. Cycle parity is therefore preserved modulo the number of NOT edges. By Harary's criterion, $G_A(I_C)$ is bipartite exactly when every source cycle has even NOT parity. The odd-NOT cycle from Lemma 4 is thus an intrinsic non-bipartite cycle in the hard liquidation instance.

Theorem 4. *There exists a constant $\delta_0 > 0$ such that, in the non-bipartite regime, finding a δ_0-approximate clearing in collateral-only linear price-impact liquidation markets is PPAD-hard. Consequently, for a sufficiently small fixed δ_0, the problem is PPAD-complete.*

Proof. Decrease the source constant κ from Lemma 4, if necessary, to a positive rational constant, and let B be the constant from Lemma 5. Choose $\delta_0 = \kappa/(10B)$. Given a promised unbalanced SUM/NOT circuit C, build I_C with no disconnected anchor. By Lemma 6, the asset borrowing graph of I_C is non-bipartite. Any δ_0-clearing decodes to a κ-satisfying assignment of C. Thus a clearing oracle for non-bipartite instances would solve a PPAD-complete source problem. Membership follows from the PPAD proposition.

The quantifiers in Theorem 4 are worst-case. We do not claim that every non-bipartite borrowing graph makes clearing hard. Rather, Lemma 6 shows that the constructed hard instances are non-bipartite for an intrinsic reason. The source circuit contains odd-NOT frustration, and the liquidation gadgets preserve that parity as an odd collateral-debt cycle. The non-bipartiteness is therefore part of the hard component, not a disconnected certificate attached after the fact.

Theorem 5 (AMM price-impact hardness). *There exists a constant $\delta_{\mathrm{cp}} > 0$ such that, in the non-bipartite regime, finding a δ_{cp}-approximate clearing remains PPAD-hard. This holds for collateral-only liquidation markets with independent constant-product marginal price curves*

$$p_k = P_0(1 + Q_k/R_k)^{-2}.$$

The same hardness holds for execution-price curves $p_k = P_0(1+Q_k/R_k)^{-1}$. Consequently, for a sufficiently small fixed δ_{cp}, constant-product smooth clearing is PPAD-complete.

Proof. Decrease the source constant κ in Lemma 4, if necessary, to a positive rational constant at most one. Build the same SUM/NOT instance as in Theorem 4, but use the lifted encoding $y_v = \Pi_{[0,1]}(P_0 - p_v)$, the writer sheet, and the constant-product pools of Appendix E, with $P_0 = 60/\kappa$ and $\delta_{\mathrm{cp}} = \kappa/120$. The collateral-debt incidences and assets are unchanged, so Lemma 6 gives the same intrinsic non-bipartite cycle. Lemmas 16–19 imply that every δ_{cp}-clearing decodes to a SUM/NOT assignment with gate violation at most $\kappa/2 < \kappa$. Thus a clearing oracle would solve the PPAD-hard promised SUM/NOT source. Membership follows from the PPAD proposition above.

Related work and discussion. The tractable linear side sits in classical Z-matrix complementarity and least-element theory [28, 26, 29–31, 33]. Our role is to identify bipartite DeFi liquidation clearing as a bounded Z-matrix complementarity problem and to show what happens when the sign condition fails. Prior clearing and fire-sale work already exposes Tarski/lattice structure [20, 21, 3], and Besting–Hoefer–Huth [1] compute extremal fixed points for generalized Eisenberg–Noe payment networks. Their setting is conservation-like payment allocation, while ours is price-impact clearing with possible super-unit feedback. Multi-asset fire-sale and collateral-network models study endogenous liquidation prices, existence, uniqueness, strategies, and contagion [12, 13, 5, 32]. Our focus is complementary bit-complexity, sign transforms, fixed-asset algorithms, and PPAD boundaries. CDS hardness uses derivative contingencies [23, 6]; here NOT comes from collateral-only debt relief, with graph balance understood in the classical sense of Harary [19]. DeFi liquidation, lending, and CPMM-oracle work supplies the economic setting [15–18, 4, 35].

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A Protocol Snapshot Calibration

To check that collateral/debt role reversal is not only a syntactic possibility, we queried Morpho's public GraphQL API [36] at

`https://api.morpho.org/graphql`

on 18 June 2026, using the `markets` query ordered by `BorrowAssetsUsd` in descending order, `first` = 1000, `skip` = 0, and `chainIds` = [1]. The table below summarizes the first 700 Ethereum markets returned by that public API, the active edge list, and an odd-cycle witness. Each Morpho market gives an ordered collateral–loan asset pair. The corresponding undirected edge is active at threshold T if that market's USD borrow value and USD collateral value are both at least T; repeated asset pairs are then naturally deduplicated in the graph. The primary witness at the 10,000-USD threshold is

$$\mathrm{LBTC} \to \mathrm{USDC}, \qquad \mathrm{WBTC} \to \mathrm{USDC}, \qquad \mathrm{WBTC} \to \mathrm{LBTC},$$

where arrows point from collateral asset to loan asset. These three directed markets give the undirected asset triangle LBTC–USDC–WBTC–LBTC. This is a snapshot-level relevance check for realized market structure; it is not used by any hardness proof.

The exact query shape was:

```
query Markets($first: Int!, $skip: Int!, $chainIds: [Int!]) {
  markets(
    first: $first
    skip: $skip
    orderBy: BorrowAssetsUsd
    orderDirection: Desc
    where: { chainId_in: $chainIds }
  ) {
    items {
      marketId
      chain { id network }
      loanAsset { address symbol decimals }
      collateralAsset { address symbol decimals }
      oracle { address }
      lltv
      state {
        borrowAssetsUsd
        collateralAssetsUsd
        supplyAssetsUsd
        liquidityAssetsUsd
      }
    }
  }
}
```

Threshold (USD)	Assets	Active edges	Bipartite?	Odd cycle
0	365	554	no	yes
100	153	214	no	yes
1,000	140	192	no	yes
10,000	122	172	no	yes

Odd-cycle witness at threshold USD 10,000: LBTC (0x8236...4494) – USDC (0xA0b8...eB48) – WBTC (0x2260...C599) – LBTC (0x8236...4494).

Scale of the small-impact condition. The snapshot above certifies role reversal, not price-impact gain. To calibrate market depth, we use a mainstream odd cycle from the same 10,000-USD threshold graph, USDC–WBTC–WETH–USDC, query Uniswap v3 pools at Ethereum block 25,345,235, and compare small collateral-sale stresses with the active virtual reserve of the collateral token. The three pools used below are WBTC/USDC at fee tier 3000, pool $0x99ac8ca7087fa4a2a1fb6357269965a2014abc35$; WBTC/WETH at fee tier 500, pool $0x4585fe77225b41b697c938b018e2ac67ac5a20c0$; and USDC/WETH at fee tier 500, pool $0x88e6a0c2ddd26feeb64f039a2c41296fcb3f5640$. For a Uniswap v3 pool with active liquidity L and $\sqrt{P} = \texttt{sqrtPriceX96}/2^{96}$, the active virtual reserves are $L/\sqrt{P}$ for token0 and $L\sqrt{P}$ for token1, adjusted by token decimals.

Collat.	Debt	Collat. USD	Reserve USD	$f_{1\%}$	$\text{Gain}_{\eta=1\%}$
WBTC	USDC	233,592,318	54,967,235	4.25%	8.5
WBTC	WETH	229,307	283,619,296	<0.01%	<0.1
WETH	USDC	6,202,247	242,634,634	0.03%	<0.1

Uniswap v3 pools sampled at Ethereum block 25345235.

The small-impact contraction condition in the theory depends on market depth and smoothing width, so this table should be read as a stress calibration. Let $f = Q/R$ be the fraction of the relevant AMM reserve sold into a constant-product pool, and let $\eta = \varepsilon/(Cp_a)$ be the liquidation smoothing width as a fraction of collateral value. For a marginal constant-product price $p(Q) = P(1 + Q/R)^{-2}$, the first-order local relative price change is approximately $2f$. For an execution-price curve $p(Q) = P(1 + Q/R)^{-1}$, it is approximately f. Thus a one-position first-order local feedback gain is on the order of $2f/\eta$ for marginal pricing and f/η for execution pricing. The unit-gain thresholds are:

η	0.5%	1%	2%	5%
marginal price: $f > \eta/2$	0.25%	0.5%	1%	2.5%
execution price: $f > \eta$	0.5%	1%	2%	5%

This calculation does not assert that a particular protocol oracle transmits AMM spot moves instantly. It explains why the large-impact regime is a realistic stress regime once smoothing is narrow relative to possible liquidation size.

B Details for Fixed-Asset-Dimension Algorithms

Proof (Details for Theorem 2). The $2m$ affine hyperplanes $h_i(p) = 0$ and $h_i(p) = \varepsilon_i$ partition $\mathbb{R}^n$ into faces. Standard fixed-dimensional arrangement algorithms enumerate all faces, their sign vectors, and rational sample points in $(m+n)^{O(n)}\,\mathrm{poly}(L)$ time [10]. On a face, each liquidation variable is fixed as 0, $h_i(p)/\varepsilon_i$, or 1. Boundary faces cause no ambiguity because the adjacent formulas agree at $h_i = 0$ and $h_i = \varepsilon_i$. Thus the pre-clip price update

$$\pi_k(p) = p_k^0 - s_k \sum_{i:a_i=k} \alpha_i x_i(p)$$

is rational affine on the face.

Adding the hyperplanes $\pi_k(p) = 0$ and $\pi_k(p) = M_k$ for all assets refines the face into at most $n^{O(n)}$ subfaces, still $(m+n)^{O(n)}$ in total. On each subface, the full map is affine, $F(p) = Ap+b$. A clearing on that subface is the rational linear feasibility problem

$$(I-A)p = b, \qquad p \in C',$$

where C' is the rational polyhedron describing the subface. Exact elimination and rational linear feasibility have polynomial bit complexity and return a rational point of polynomial bit length when feasible. Since F is continuous on the compact price box, Brouwer guarantees that at least one subface is feasible.

Proof (Details for Theorem 3). The liquidation arrangement is the same as in Theorem 2. On each face, $Q_k(p) = \sum_{i:a_i=k} \alpha_i x_i(p)$ is rational affine. For an execution-price constant-product pool, the uncapped branch is the quadratic equation

$$p_k(R_k + Q_k(p)) = \kappa_k,$$

while the capped branch is $p_k = M_k$ together with $\kappa_k \geq M_k(R_k + Q_k(p))$; the uncapped branch uses the reverse inequality. Bounded-degree rational curves are handled by clearing denominators, which are assumed bounded away from zero on the relevant sales range. For each face and each of the 2^n cap choices, fixed because n is the dimension parameter, we obtain a bounded-degree semialgebraic system in n variables.

Renegar's fixed-dimensional algorithms, or the Basu–Pollack–Roy roadmap algorithms, decide such systems and produce rational approximations to real solutions in time polynomial in the coefficient bit length and $\log(1/\eta)$ [22, 2]. For constant-product pools, $R_k + Q_k(p) \geq R_k > 0$, and for the stated rational curves the denominators have explicit positive lower bounds; hence F has an effective Lipschitz bound K_F of polynomial bit length on the compact box. If $\|\hat{p} - p^*\|_\infty \leq \eta$ for an exact fixed point p^* and $\hat{x} = x(\hat{p})$, then the liquidation equations hold exactly and

$$\|\hat{p} - F(\hat{p})\|_\infty \leq (1 + K_F)\eta.$$

Taking $\eta \leq \delta/(1+K_F)$ gives a δ-clearing. Enumerating all faces and cap branches gives the claimed running time.

C Additional Details for Sign-Balanced Clearing

This appendix records mechanical extensions and algebraic details used around the sign-balanced side.

Proposition 4 (same-asset linear debt buy pressure). *The sign-balanced tractability result is robust to a separable linear buy-pressure term on the debt asset. Replace* (3) *by*

$$p_k = \Pi_{[0,M_k]}\left(p_k^0 - s_k^- \sum_{i:a_i=k} \alpha_i x_i + s_k^+ \sum_{i:b_i=k} \beta_i x_i\right),$$

with $s_k^-, s_k^+, \alpha_i, \beta_i \geq 0$. If G_A is bipartite, the same transform as in Lemma 1 again gives $y = \Pi_{[0,u]}(Ay+c)$ with $A \geq 0$. Hence the exact polynomial-time linear result and the monotone CLS upper bound continue to hold for this extension.

Proof (Proof of Proposition 4). Only the price block changes. If $k \in L$, then $q_k = p_k$. A collateral-sale term with $a_i = k$ has $x_i = 1 - z_i$, contributing a positive coefficient $s_k^- \alpha_i z_i$ after absorbing the constant. A debt-buy term with $b_i = k$ has $a_i \in R$, hence $x_i = z_i$, and contributes the positive coefficient $s_k^+ \beta_i z_i$.

If $k \in R$, then $q_k = M_k - p_k = \Pi_{[0,M_k]}(M_k - r_k)$, where r_k is the unclipped price expression. A collateral-sale term with $a_i = k$ has $x_i = z_i$ and becomes $+s_k^- \alpha_i z_i$ in $M_k - r_k$. A debt-buy term with $b_i = k$ has $a_i \in L$, hence $x_i = 1 - z_i$, and the term $-s_k^+ \beta_i (1 - z_i)$ contributes a positive coefficient $s_k^+ \beta_i z_i$ after absorbing the constant. The liquidation block is unchanged from Lemma 1. Thus all variable coefficients remain nonnegative.

Proposition 5 (small-impact contraction). *In the transformed bipartite system $T(y) = \Pi_{[0,u]}(Ay + c)$, suppose there are rational $v > 0$ and $\lambda < 1$ with $Av \leq \lambda v$. Then the clearing is unique, and Banach iteration computes a δ-clearing in time polynomial in the input size, the bit length of (v, λ), $\log(1/\delta)$, and $1/(1-\lambda)$. In particular this is polynomial whenever such a contraction certificate has polynomial bit length and inverse-polynomial gap.*

Proof (Proof of Proposition 5). In the weighted sup norm $\|r\|_v = \max_i |r_i|/v_i$, the projection $\Pi_{[0,u]}$ is nonexpansive coordinate-wise and, using the assumed certificate $Av \leq \lambda v$,

$$\|T(y) - T(y')\|_v \leq \|A(y - y')\|_v \leq \lambda \|y - y'\|_v.$$

Thus T is a contraction, so Banach iteration converges geometrically to the unique fixed point. The number of iterations is $O(\log(R/\delta)/(1-\lambda))$, where R is the box diameter in $\|\cdot\|_v$.

C.1 Nonnegative-Drift Special Case

Lemma 7 (the least fixed point has subcritical interior). *For any $B \geq 0$ and any $g \in \mathbb{R}^n$, the least fixed point of $T(z) = \Pi_{[0,u]}(Bz + g)$ satisfies*

$$\rho(B_{I^* I^*}) < 1.$$

Proof. This is the fixed-point form of the standard least-element fact that the basic block of a least solution of a Z-matrix complementarity problem is a nonsingular M-matrix [26, 29, 33]. For completeness we give the short monotone proof. Suppose $M = B_{I^*I^*}$ has $\rho(M) \geq 1$. Perron–Frobenius gives a nonzero $\delta \geq 0$ with $M\delta = \rho(M)\delta$, allowing reducibility. Extend δ by zero outside I^*. For sufficiently small $\varepsilon > 0$, $z(\varepsilon) = z^* - \varepsilon\delta$ remains in the same open box on I^*. On I^*,

$$(Bz(\varepsilon) + g)_{I^*} = z^*_{I^*} - \varepsilon M\delta \leq z^*_{I^*} - \varepsilon\delta = z(\varepsilon)_{I^*}.$$

On U^*, decreasing z^* in nonnegative directions can only decrease the drive, so clipping gives $T_i(z(\varepsilon)) \leq u_i = z_i(\varepsilon)$. On L^*, the fixed-point condition implies $(Bz^* + g)_i \leq 0$, and the drive again only decreases, so $T_i(z(\varepsilon)) = 0 = z_i(\varepsilon)$. Thus $T(z(\varepsilon)) \leq z(\varepsilon)$. The least fixed point of a monotone map is the least prefixed point, so $z^* \leq z(\varepsilon) < z^*$ on the support of δ, a contradiction.

Lemma 8 (floor equals unreachability). *For $g \geq 0$, the positive coordinates of the least fixed point are exactly those reachable from* $\operatorname{supp}(g)$ *in the feeds graph $j \to i$ when $B_{ij} > 0$. The restriction to this reachable set is autonomous and strictly positive.*

Proof (Proof of Lemma 8). Let A be the set of coordinates reachable from $\operatorname{supp}(g)$ in the feeds graph, and L its complement. In the Kleene iteration from 0, a coordinate becomes positive exactly when it receives positive input from g or from an already positive coordinate, since all coefficients and all existing positive values are nonnegative. Hence the eventual positive coordinates are exactly A. If a coordinate in A fed a coordinate in L, then the latter would also be reachable. Therefore the restricted active block on A is autonomous.

Lemma 9 (nonnegative cascade). *An autonomous block with $B \geq 0$, $g \geq 0$, and every coordinate reachable from* $\operatorname{supp}(g)$ *has a unique fixed point, computable in $O(n)$ cascade rounds by exact rational linear solves and standard exact M-matrix tests.*

Proof (Proof of Lemma 9). This is a nonnegative-drift specialization of the least-element Z-matrix methods discussed in Section 3.1. We give the fixed-point proof because it is useful for interpreting the economic special case below. Write S for the block. By Lemma 8, $z^* > 0$, so $I^* = \{i : z^*_i < u_i\} = S \setminus U^*$. By Lemma 7, $\rho(B_{I^*I^*}) < 1$. Maintain a tentative clamp set C, initially empty, put $I = S \setminus C$, and when $\rho(B_{II}) < 1$ write

$$\hat{z}_I(C) = (I - B_{II})^{-1}(g_I + B_{IC}u_C).$$

One round executes the first applicable case:

- (G1) if $\rho(B_{II}) \geq 1$: move any one coordinate of I into C;
- (G2) else if $\hat{z}_i(C) \geq u_i$ for some $i \in I$: move all such i into C;
- (R) else form z with $z_C = u_C$, $z_I = \hat{z}_I(C)$, and release bad $= \{c \in C : (Bz + g)_c < u_c\}$, halting if this set is empty.

Growth terminates because enlarging C only shrinks the principal submatrix B_{II}. Every repair point is prefixed. Indeed, $z_I = \hat{z}_I(C) \geq 0$ by the Neumann series, $(Bz + g)_I = z_I$, and $T(z)_C \leq u_C = z_C$, so $z \geq z^*$. Thus a released coordinate cannot be in the true clamp set U^*. After the first repair we have $C \supseteq U^*$, hence the next interior $I \subseteq I^*$ is subcritical. The old repair point is prefixed for the newly released subproblem, so the next solve is coordinate-wise no larger. Released coordinates stay below u, no re-clamping occurs, and C strictly shrinks on each later repair. Hence the cascade halts after at most $2|S|$ rounds.

At halt $T(z) = z$. If $z' \geq z^*$ is another fixed point and $D = \{i : z'_i > z^*_i\}$, then $D \subseteq I^*$ and $\delta_D \leq B_{DD}\delta_D$ for $\delta = z' - z^* > 0$ on D. Collatz–Wielandt gives $\rho(B_{DD}) \geq 1$, contradicting $D \subseteq I^*$ and $\rho(B_{I^*I^*}) < 1$. Thus the halted fixed point is unique and equals z^*.

Theorem 6 (nonnegative-drift SFBFP). *For every rational $B \geq 0$, $g \geq 0$, and $u > 0$, the least fixed point of $z = \Pi_{[0,u]}(Bz + g)$ is computable in time polynomial in the input bit length.*

Proof (Proof of Theorem 6). The result is a special case of Proposition 2. The following argument is included as a direct monotone proof for the nonnegative-drift regime. Compute the reachable set A of Lemma 8 by graph search and set $z^*_L = 0$. The active subsystem on A is autonomous and satisfies the hypotheses of Lemma 9, so the cascade returns its unique fixed point in $O(|A|)$ rounds. The exact linear solves have polynomial bit complexity, and each subcriticality decision is a standard exact rational M-matrix test. Combining the active solution with $z^*_L = 0$ gives the least fixed point of the original system.

C.2 A Self-Contained Activation Cascade

The polynomial-time result in Proposition 2 follows from classical least-element complementarity theory. We nevertheless record the monotone-map-native cascade that specializes those least-element methods to the fixed-point form.

Lemma 10 (from-above cascade for an active block with positive least fixed point). *Let $A \subseteq S$ carry an autonomous block map $T_A(y) = \Pi_{[0,u_A]}(B_{AA}y + g_A)$ with $B_{AA} \geq 0$ and* arbitrary *g_A, and suppose its least fixed point $\zeta = \mathrm{lfp}(T_A)$ is strictly positive. Then ζ is the unique fixed point of T_A. It is computed using at most $|A|$ exact linear solves and at most $|A| + 1$ rounds by the from-above cascade, which starts with all coordinates clamped ($C = A$). In a round with $I = A \setminus C$, the algorithm solves $\hat{z}_I = (I - B_{II})^{-1}(g_I + B_{IC}u_C)$. It clamps every $i \in I$ with $\hat{z}_i \geq u_i$. If no such coordinate exists, it releases from C every c with $(Bz + g)_c < u_c$, and halts when none remain.*

Proof. Since $\zeta > 0$, Lemma 7 applied to T_A gives $\rho(B_{I_A I_A}) < 1$, where $I_A = \{i \in A : \zeta_i < u_i\}$ and $U_A = \{i \in A : \zeta_i = u_i\}$.

Uniqueness. Any fixed point z' has $\delta = z' - \zeta \geq 0$. If $\delta_i > 0$, then $\zeta_i < u_i$, and since $\zeta > 0$, $i \in I_A$. Hence $\operatorname{supp}\delta \subseteq I_A$. On $D = \operatorname{supp}\delta$, interiority gives $\delta_D \le B_{DD}\delta_D$. Collatz–Wielandt then forces $\rho(B_{DD}) \ge 1$, contradicting $\rho(B_{DD}) \le \rho(B_{I_A I_A}) < 1$ unless $\delta = 0$.

Invariant $C \supseteq U_A$. Initially $C = A \supseteq U_A$. Each repair point satisfies $z \ge \zeta$, as shown below. A released c, with $(Bz+g)_c < u_c$, then has $(B\zeta+g)_c \le (Bz+g)_c < u_c$. Thus $\zeta_c < u_c$ and $c \notin U_A$. Hence $C \supseteq U_A$ throughout and $I = A \setminus C \subseteq I_A$. The interior is always subcritical, and $(I - B_{II})^{-1} \ge 0$.

Positivity by domination. For $C \supseteq U_A$, the true ζ satisfies $\zeta_I = (I-B_{II})^{-1}(g_I + B_{IC}\zeta_C)$. Therefore $\hat{z}_I - \zeta_I = (I - B_{II})^{-1}B_{IC}(u_C - \zeta_C) \ge 0$, and $\hat{z}_I \ge \zeta_I > 0$. At a repair round, where $\hat{z}_I < u_I$, the point z equals u on C and $\hat{z}_I \in (0, u_I)$ on I. It is prefixed because $T_A(z)_I = \hat{z}_I = z_I$ and $T_A(z)_C \le u_C = z_C$. Hence $z \ge \zeta$.

Monotone repair, no re-clamping. From $C = A$, the first round is a repair. If a repair round has a nonempty release set bad, put $C' = C \setminus \text{bad} \supseteq U_A$ and $I' = A \setminus C' \subseteq I_A$. The contraction $T_{C'}(w) = \Pi_{[0,u_{I'}]}(B_{I'I'}w + B_{I'C'}u_{C'} + g_{I'})$ has a unique fixed point $\hat{z}_{I'}(C')$. The current $z|_{I'}$ is prefixed for it, with equality on the old interior and strict decrease on bad. Hence $\hat{z}_{I'}(C') \le z_{I'}$, so $\hat{z}_c < u_c$ on bad and $\hat{z}_i < u_i$ on the old interior. The next round is again a repair with $C' \subsetneq C$. After the first repair, each round removes at least one coordinate, so within $\le |A|$ rounds the release set is empty and the returned fixed point equals ζ.

Lemma 11 (monotone activation). *Start from $A = \varnothing$, $z = 0$. With $d = Bz + g$, let $\text{act} = \{f \notin A : d_f > 0\}$. If $\text{act} = \varnothing$, stop. Otherwise set $A \leftarrow A \cup \text{act}$, recompute $z_A = \text{lfp}(T_A)$ by Lemma 10, set $z = 0$ off A, and repeat. This runs for at most $|S|$ iterations, every block it solves has $\text{lfp}(T_A) > 0$, and it returns the least fixed point z^*.*

Proof. Maintain the following invariant. We have $A \subseteq A^* := \text{supp}(z^*)$, $z = \text{lfp}(T_A)$ on A and 0 off A, $\text{lfp}(T_A) > 0$, and $z \le z^*$. The base case $A = \varnothing$ is vacuous.

Activations are correct. If $d_f = (Bz+g)_f > 0$ with $z \le z^*$, then $(Bz^*+g)_f \ge d_f > 0$. Thus $z^*_f > 0$, so $f \in A^*$ and $A \cup \text{act} \subseteq A^*$.

The new block's least fixed point is positive. Write $A' = A \cup \text{act}$. Activation only adds nonnegative input, so the Kleene iterates from 0 for $T_{A'}$ dominate those for T_A on A. Hence $\text{lfp}(T_{A'})|_A \ge \text{lfp}(T_A) > 0$. For $f \in \text{act}$, $(B\,\text{lfp}(T_{A'}) + g)_f \ge d_f > 0$, so that coordinate is positive too. Thus $\text{lfp}(T_{A'}) > 0$ and Lemma 10 applies.

$z \le z^*$ is preserved. $z^*_{A'} = \Pi\big(B_{A'A'}z^*_{A'} + B_{A',S\setminus A'}z^*_{S\setminus A'} + g_{A'}\big) \ge \Pi(B_{A'A'}z^*_{A'} + g_{A'}) = T_{A'}(z^*_{A'})$, so $z^*_{A'}$ is prefixed for $T_{A'}$ and $\text{lfp}(T_{A'}) \le z^*_{A'}$.

Termination and correctness. Each productive iteration enlarges A, so there are at most $|S|$. At termination $d_f \le 0$ off A, so $T(z)_f = \Pi(d_f) = 0 = z_f$. On A, z is a fixed point of T_A. Thus $T(z) = z$ and $z \le z^*$. Since z^* is least, $z = z^*$.

Proposition 6 (when the transformed drift is nonnegative). *Suppose G_A is bipartite. The transformed drift c of Lemma 1 can be made nonnegative exactly in the two-sector case in which no asset is used both as collateral and as debt, the bipartition is chosen with all collateral assets in L and all debt assets in R,* *and*

$$p_k^0 \geq s_k \sum_{i:a_i=k} \alpha_i \quad (k \in L), \qquad p_k^0 \leq M_k \quad (k \in R), \qquad \varepsilon_i \geq \theta_i D_i M_{b_i} \quad (i \in [m]).$$

In that regime the clearing is computable in polynomial time by Theorem 6, even when the price impact is large ($\rho(A) \geq 1$).

Proof (Proof of Proposition 6). With all collateral assets in L and all debt assets in R, the four cases of Lemma 1 give

$$c_k = \begin{cases} p_k^0 - s_k \sum_{i:a_i=k} \alpha_i, & k \in L, \\ M_k - p_k^0, & k \in R, \end{cases} \qquad c_i = 1 - \frac{\theta_i D_i M_{b_i}}{\varepsilon_i}.$$

The displayed inequalities are exactly nonnegativity of these entries. If an asset is used both as collateral and as debt, then under any bipartition one witnessing position has $a_i \in R$, and Lemma 1 gives the strictly negative drift $-C_i M_{a_i}/\varepsilon_i$. Hence no bipartition can make $c \geq 0$.

Remark 3. The two-sector condition is strictly stronger than bipartiteness. It forbids the asset role-reversal that creates the debt-relief feedback of Fig. 1. Theorem 6 is therefore a useful special case of the general algorithm, not the boundary result.

Gauge normal form for balanced SUM/NOT.

Lemma 12 (gauge normal form). *Let $y_v = \Pi_{[0,1]}(y_u + y_w)$ be a truncated-SUM gate and write $\bar{t} = 1 - t$. Then*

$$\bar{y}_v = 1 - \Pi_{[0,1]}(y_u + y_w) = \max\{1 - y_u - y_w, 0\} = \Pi_{[0,1]}(\bar{y}_u + \bar{y}_w - 1).$$

Consequently, applying $y \mapsto 1 - y$ to a Harary-balanced subset of wires turns a SUM/NOT circuit into a subtraction-free clipped-affine system $y = \Pi_{[0,1]}(Ay+c)$ with $A \geq 0$.

Proof. The identity uses $1 - \Pi_{[0,1]}(t) = \Pi_{[0,1]}(1-t)$ on the relevant range. A balanced signed dependency graph admits a vertex flip making every edge positive. Performing $y \mapsto 1 - y$ on the flipped wires replaces each NOT by an identity and each SUM by a gate of the displayed form, whose variable coefficients are nonnegative.

Corollary 3 (balanced circuit fragment). *The sign-balanced SUM/NOT fragment is the circuit analogue of the transformed DeFi system in Lemma 1: after a gauge transform it has the form $y = \Pi(Ay + c)$ with $A \geq 0$. Therefore the balanced fragment is polynomial-time solvable by Proposition 2; SUM/NOT hardness must come from sign-imbalance.*

Separable monotone impact and comparative statics.

Theorem 7 (separable monotone price impact). *Replace the linear price equation by*

$$p_k = g_k\left(\sum_{i:a_i=k} \alpha_i x_i\right),$$

where each $g_k : [0,\infty) \to [0, M_k]$ is continuous, nonincreasing, and given with an exact rational evaluation algorithm: on every rational input of polynomial bit length in the relevant sales range, it outputs the exact rational value $g_k(Q)$ in polynomial time and polynomial bit length. If G_A is bipartite, then for every rational $\delta > 0$, smooth δ-clearing reduces in polynomial time to succinct Tarski and hence belongs to CLS. This includes independent asset–numeraire constant-product pools such as $g_k(Q) = \min\{\kappa_k/(R_k + Q), M_k\}$.

Proof (Proof of Theorem 7). Use the same bipartition (L, R) and transformed variables q, z as in Theorem 1. For the price block, if $k \in L$, then $x_i = 1 - z_i$ for positions collateralized by k, so the sale quantity is nonincreasing in each z_i. Since g_k is nonincreasing, $q_k = p_k$ is nondecreasing. If $k \in R$, then $x_i = z_i$, the sale quantity is nondecreasing, $p_k = g_k(\cdot)$ is nonincreasing, and $q_k = M_k - p_k$ is nondecreasing. The liquidation block is the same four-case computation as Lemma 1, so the transformed map is isotone. On a rational grid point, the exact-evaluation assumption gives every coordinate of $T(y)$ as a rational of polynomial bit length; therefore the coordinate-wise floor to the rational grid can be computed by exact rational comparison, with no boundary ambiguity. The grid-rounding proof of Theorem 1 applies.

Proof (Proof of Proposition 1). Use the sign transform of Theorem 1. For an asset $k \in L$, write

$$Q_k(z) = \sum_{i:a_i=k} \alpha_i x_i = Q_k^0 - \sum_{i:a_i=k} \alpha_i z_i, \qquad Q_k^0 = \sum_{i:a_i=k} \alpha_i,$$

and set $t_k = \sum_{i:a_i=k} \alpha_i z_i$, $h_k(t) = g_k(Q_k^0 - t)$. For $k \in R$, set $t_k = \sum_{i:a_i=k} \alpha_i z_i$ and $h_k(t) = M_k - g_k(t)$. In both cases h_k is a continuous nondecreasing rational PWL function of the nonnegative affine input t_k, and the transformed price variable is $q_k = h_k(t_k)$.

Let h be any such nondecreasing PWL function with breakpoints

$$0 = \tau_0 < \tau_1 < \cdots < \tau_r$$

on its relevant interval and slope $s_j \geq 0$ on $[\tau_{j-1}, \tau_j]$. Then

$$h(t) = h(0) + \sum_{j=1}^{r} s_j \, \Pi_{[0,\tau_j-\tau_{j-1}]}\big(t - \tau_{j-1}\big).$$

For every price curve segment introduce an auxiliary variable

$$w_{kj} = \Pi_{[0,\tau_j-\tau_{j-1}]}\big(t_k - \tau_{j-1}\big).$$

The equations for the w_{kj}'s and the transformed prices

$$q_k = \Pi_{[0,M_k]}\left(h_k(0) + \sum_j s_{kj} w_{kj}\right)$$

are clipped-affine equations with nonnegative variable coefficients; the outer clip is harmless because h_k already ranges in $[0, M_k]$. Adding them to the transformed liquidation equations gives a polynomial-size system

$$Y = \Pi_{[0,U]}(A'Y + c'), \qquad A' \geq 0,$$

with rational data. Proposition 2 computes its least fixed point exactly in polynomial time. Projecting this solution to the original price and liquidation variables gives the least sign-balanced clearing for the PWL-impact instance.

Proposition 7 (sign-balanced comparative statics). *Fix a grid used in the proof of Theorem 1. Suppose two sign-balanced instances have transformed grid maps T_η and S_η with $T_\eta(y) \leq S_\eta(y)$ for every grid point y. Then the least grid clearing of T_η is no larger than the least grid clearing of S_η, and the same monotone order holds for the greatest grid clearings.*

Proof. On a finite lattice, the least fixed point of an isotone map is obtained by iterating from the bottom element, and the greatest fixed point by iterating from the top. The pointwise inequality and isotonicity imply the comparison by induction.

Proposition 8 (Lipschitz tractability). *In the separable monotone-impact model of Theorem 7, suppose each g_k is γ_k-Lipschitz on the relevant sales range. Form the nonnegative gain matrix*

$$\widehat{A} = \begin{pmatrix} 0 & \widehat{P} \\ \widehat{Q} & 0 \end{pmatrix}, \qquad \widehat{P}_{ki} = \gamma_k \alpha_i \ (a_i = k), \quad \widehat{Q}_{ia_i} = \frac{C_i}{\varepsilon_i}, \quad \widehat{Q}_{ib_i} = \frac{\theta_i D_i}{\varepsilon_i}.$$

If there are rational $v > 0$ and $\lambda < 1$ with $\widehat{A}v \leq \lambda v$, then the transformed clearing map is a contraction in the weighted sup norm induced by v. The clearing is unique, and fixed-point iteration computes a δ-clearing in time polynomial in the input size, the bit length of (v, λ), $\log(1/\delta)$, and $1/(1-\lambda)$. In particular this is polynomial whenever the contraction certificate has polynomial bit length and inverse-polynomial gap.

Proof. The clipping maps are coordinate-wise nonexpansive, and the transformed price and liquidation blocks are Lipschitz-bounded by $\widehat{P}$ and $\widehat{Q}$, so $|T(y) - T(y')| \leq \widehat{A}|y - y'|$. With the displayed certificate, the weighted norm $\|r\|_v = \max_i |r_i|/v_i$ has Lipschitz factor at most λ. Thus $O(\log(R/\delta)/(1-\lambda))$ iterations suffice, where R is the diameter of the rational box in this norm.

Box-LCP and exact subcriticality tests.

Lemma 13 (algebraic box-LCP form). *For $B \geq 0$ and any g, the least fixed point z^*, its floor set L^*, and its clamp set U^* are described by a finite system of linear equations and rational inequalities:*

$$z^*_{L^*} = 0, \qquad z^*_{U^*} = u_{U^*}, \qquad z^*_{I^*} = (I - B_{I^*I^*})^{-1}\big(g_{I^*} + B_{I^*U^*}u_{U^*}\big),$$

together with the lower-, interior-, and upper-bound inequalities defining the three sets.

Proof. This is the usual active-set description of a box complementarity solution. On I^*, the clipping is inactive, and the inverse exists by Lemma 7. The boundary inequalities are exactly the box complementarity conditions.

Lemma 14 (exact subcriticality tests). *For rational $B' \geq 0$, the condition $\rho(B') < 1$ can be tested by exact rational polynomial-time M-matrix tests for $I - B'$. No numerical eigenvalue approximation or transcendence bound is needed.*

Proof. $I - B'$ is a Z-matrix, and $\rho(B') < 1$ iff $I - B'$ is a nonsingular M-matrix iff there is $v > 0$ with $(I - B')v > 0$, by the standard Fiedler–Ptak and nonnegative-matrix characterizations [25, 24]. After normalization, this is a rational linear feasibility problem; equivalent principal-minor or elimination tests give the same exact decision.

Lemma 15 (subcritical upper bound). *Assume $\rho(B_{SS}) < 1$ and $g' \geq 0$. The unclipped solve $\hat{z} = (I - B_{SS})^{-1}g'$ dominates the least box fixed point on S. If $\hat{z} \leq u_S$, then $\hat{z}$ is exact on S. Otherwise, the over-threshold set can strictly contain the true clamp set.*

Proof. This is the standard monotone comparison for a subcritical M-matrix block. The clipped Kleene orbit from 0 is bounded above by the unclipped affine orbit, which increases to $\hat{z}$. Strict over-capping occurs for

$$B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad g' = (0, 10), \qquad u = (5, 1),$$

where $\hat{z} = (10, 10)$ but the box fixed point is $(1, 1)$.

For $y = \Pi_{[0,u]}(Ay + c)$, define $B = I - A$ and $d = -c$. Then $F(y) = By + d$. Coordinate-wise, the fixed-point condition is equivalent to

$$y_i = 0 \Rightarrow F_i(y) \geq 0, \quad 0 < y_i < u_i \Rightarrow F_i(y) = 0, \quad y_i = u_i \Rightarrow F_i(y) \leq 0.$$

Introducing an upper-bound multiplier $w \geq 0$ gives the upper-bounded LCP formulation

$$0 \leq y \perp By + d + w \geq 0, \qquad 0 \leq w \perp u - y \geq 0.$$

Indeed, if $0 < y_i < u_i$, then $w_i = 0$ and $(By+d)_i = 0$. If $y_i = 0$ and $u_i > 0$, then $w_i = 0$ and $(By + d)_i \geq 0$. If $y_i = u_i$, then $(By + d)_i = -w_i \leq 0$. Coordinates with $u_i = 0$ may be eliminated in preprocessing.

The ordinary standard-LCP embedding of this upper-bounded system, with variables (y, w), has matrix

$$\begin{pmatrix} B & I \\ -I & 0 \end{pmatrix}.$$

The $+I$ block is forced by the sign of the upper-bound multiplier and gives positive off-diagonal entries. Hence the bounded system should not be described as an ordinary standard-form Z-matrix LCP after this embedding. It is instead a bounded Z-matrix complementarity problem, the setting of the least-element theory used in Proposition 2. Small impact is the special case where $\rho(A) < 1$, equivalently $I - A$ is a nonsingular M-matrix and the fixed point is unique.

D Gadget Algebra and Error Bounds

For exact gadget algebra write the raw wire value as $y_v = 2 - p_v$. For an approximate clearing, public wires are decoded by

$$\widehat{y}_v = \Pi_{[0,1]}(2 - p_v).$$

The output-price residual and the constraint $x_v \in [0, 1]$ imply that the raw value $2 - p_v$ is within $O(\delta)$ of $[0, 1]$. Projection onto $[0, 1]$ is nonexpansive, so this replacement only changes the local gate-error bounds by a universal constant factor. For a writer with collateral O, debt U, and parameters C, D, ε, the raw equation is

$$x = \Pi \left(\frac{2D - 2C}{\varepsilon} - \frac{D}{\varepsilon} y_U + \frac{C}{\varepsilon} y_O \right).$$

If it is the unique writer of O, then $y_O = x$ in the exact system. If $C/\varepsilon < 1$, the right-hand side is a contraction in y_O, and a δ-clearing gives an output error at most a constant multiple of $\delta/(1 - C/\varepsilon)$, including the price residual, liquidation residual, and projection step.

For the NOT gadget, $C = 1, D = 2, \varepsilon = 3$. Hence

$$y_O = \Pi \left(\frac{2 - 2y_U + y_O}{3} \right),$$

whose unique fixed point is $1 - y_U$. The self coefficient is $1/3$, so the raw residual amplification for a fixed input is at most 3.

For the average gadget, first maintain complement wires $\bar{U}, \bar{W}$ by NOT gadgets. Create an average asset A with two co-writers, both with collateral A, each contributing $\eta = 1/2$ units of output price drop. Writer 1 reads $\bar{U}$, writer 2 reads $\bar{W}$, and both use $C = 1, D = 2, \varepsilon = 3$. Since $y_{\bar{U}} = 1 - y_U$, writer 1 satisfies

$$x_1 = \frac{2}{3} y_U + \frac{1}{3} y_A,$$

and similarly $x_2 = \frac{2}{3}y_W + \frac{1}{3}y_A$. The output equation is $y_A = (x_1 + x_2)/2$, so $y_A = (y_U + y_W)/2$. The self coefficient is again bounded away from one, and the error is a constant multiple of δ.

The final gain-2 clipped writer reads $\bar{A}$ and uses $C = 1, D = 2, \varepsilon = 2$. Since $p_{\bar{A}} = 2 - (1 - y_A) = 1 + y_A$, the writer equation becomes

$$y_O = \Pi\left(y_A + \frac{1}{2}y_O\right).$$

If $y_A \leq 1/2$, the unique fixed point is $2y_A$. If $y_A > 1/2$, the unique fixed point is 1. Thus the output is $T_{[0,1]}(2y_A) = T_{[0,1]}(y_U + y_W)$. In this linear sheet all self coefficients are at most $1/2$ or $1/3$ inside the macro.

For a δ-clearing, absorb the price residual, liquidation residual, and the non-expansive projection into the following conservative local budget:

$$e_{\neg,\mathrm{dec}} \leq 5\delta, \qquad e_{A,\mathrm{raw}} \leq 7\delta, \qquad e_{\bar{A},\mathrm{raw}} \leq 10\delta,$$

$$e_{+,\mathrm{raw}} \leq 24\delta, \qquad e_{+,\mathrm{dec}} \leq 25\delta.$$

The first bound uses the self coefficient $1/3$. The average bound combines two complement inputs and the $1/3$ average self coefficient. The $\bar{A}$ bound is another NOT step, and the final gain-2 writer has self coefficient $1/2$. These constants are deliberately loose and include one extra δ for the final projection. Taking

$$B = 30$$

dominates both the NOT and SUM macro errors, so Lemma 5 holds with this explicit constant. In the lifted constant-product sheet of Appendix E, the NOT/average self coefficient is $< 1/2$ and the gain-2 self coefficient is $< 2/3$; the corresponding locked constants are $B = 30$ and $B' = 15$.

The close-factor bound can be met by scaling the output impact. For a writer with desired output contribution η, take collateral amount parameter $C = 1$, liquidation amount $\alpha = \varphi/2$, and slope $s_O = 2\eta/\varphi$. Then $s_O\alpha = \eta$ and $\alpha \leq \varphi C$.

E Explicit Constants for Constant-Product Hardness

This appendix proves Theorem 5. We use the lifted encoding $y_v = P_0 - p_v$, with the intended wire interval $[0,1] \subset [P_0 - 1, P_0]$, and write $\beta = 1/P_0$. For a writer with collateral O, debt U, parameters (C, D, ε), and $\theta = 1$, the health equation gives

$$x = \Pi_{[0,1]}\left(\underbrace{\frac{P_0(D-C)}{\varepsilon}}_{c} - \underbrace{\frac{D}{\varepsilon}}_{a}\, y_U + \underbrace{\frac{C}{\varepsilon}}_{\lambda}\, y_O\right).$$

The bias now scales with P_0. We therefore retune the writer sheet so that $c = a$, which cancels the lifted bias, while preserving the intended input gain after solving the self-dependence.

Lemma 16 (lifted parameter sheet). *With $\theta = 1$, the NOT and average co-writers use*

$$(C, D, \varepsilon) = (P_0 - 1, P_0, 2P_0 - 1),$$

and the gain-2 writer uses

$$(C, D, \varepsilon) = (2(P_0 - 1), 2P_0, 3P_0 - 2).$$

Both satisfy $c = a$. The first sheet has $a = 1-\lambda$, and the second has $a = 2(1-\lambda)$. Thus the linearized fixed points are $y_O = 1 - y_U$, $y_A = (y_U + y_W)/2$, and $y_O = \Pi_{[0,1]}(2y_A)$. The self coefficients are

$$\lambda = \frac{P_0 - 1}{2P_0 - 1}, \qquad \lambda_2 = \frac{2(P_0 - 1)}{3P_0 - 2},$$

so the residual amplification factors are $1/(1-\lambda) = 2-\beta$ and $1/(1-\lambda_2) = 3-2\beta$. The input gains are 1 for NOT, 1/2 per input for the average, and 2 for gain-2. The collateral-debt incidences are unchanged from Appendix D, so Lemma 6 applies verbatim.

Proof. For the first sheet,

$$c = \frac{P_0}{2P_0 - 1} = a, \qquad \lambda = \frac{P_0 - 1}{2P_0 - 1}, \qquad a + \lambda = 1.$$

Solving $y_O = c - ay_U + \lambda y_O$ gives $y_O = 1 - y_U$. In an average pool, two such co-writers contribute half a unit of price drop each, so the fixed point is $y_A = (y_U + y_W)/2$. For the second sheet,

$$c = \frac{2P_0}{3P_0 - 2} = a, \qquad \lambda_2 = \frac{2(P_0 - 1)}{3P_0 - 2}, \qquad a = 2(1 - \lambda_2).$$

Reading the complement wire $\bar{A}$ gives the active-branch equation $y_O = 2(1 - \lambda_2)y_A + \lambda_2 y_O$, which is equivalent to $y_O = 2y_A$, clipped at one. The displayed amplification factors follow from direct algebra after substituting $\beta = 1/P_0$.

Lemma 17 (constant-product curvature). *Price asset O by the marginal curve $p_O = P_0(1 + Q_O/R_O)^{-2}$, with $Q_O = \alpha_O x$ and slope match $\alpha_O/R_O = \beta/2$ for one writer or $\alpha_O/R_O = \beta/4$ for a two-writer average pool over range $x_1+x_2 \in [0, 2]$. Alternatively, price it by the execution curve $p_O = P_0(1+Q_O/R_O)^{-1}$, with $\alpha_O/R_O = \beta$ for one writer or $\alpha_O/R_O = \beta/2$ for an average pool. In every case,*

$$|y_O - x| \le \beta, \qquad \left| y_A - \frac{x_1 + x_2}{2} \right| \le \beta.$$

Proof. For the marginal curve, set $u = (\alpha_O/R_O)x$. With target linear price $\ell = P_0(1 - 2u)$,

$$(1 + u)^{-2} - (1 - 2u) = \frac{u^2(3 + 2u)}{(1 + u)^2} \le 4u^2.$$

In the one-writer case $u \leq \beta/2$; in the averaging pool the chosen slope gives the same bound over total range at most 2. Hence $|p_O - \ell| \leq P_0\beta^2 = \beta$. For the execution curve, with target $\ell = P_0(1-u)$,

$$(1+u)^{-1} - (1-u) = \frac{u^2}{1+u} \leq u^2.$$

Now $u \leq \beta$ in both the one-writer and average cases, so again $|p_O - \ell| \leq P_0\beta^2 = \beta$. Translating prices to $y = P_0 - p$ gives the displayed bounds.

Remark 4 (close factor). The slope matches above are compatible with the close-factor restriction. For any target ratio $\alpha/R = \rho$, choose a rational collateral sale amount $\alpha \leq \varphi C$ and set $R = \alpha/\rho$. The reserves are positive rational numbers, and changing R does not alter the collateral-debt graph.

Lemma 18 (range). *With cap $M_O = P_0$, each pool's exact price drop over the relevant gate range is less than one:*

$$\frac{1}{1+\beta} < 1 \quad \textit{for execution pricing,} \qquad \frac{1+\beta/4}{(1+\beta/2)^2} < 1 \quad \textit{for marginal pricing.}$$

Moreover, with the final choice $\delta_{\text{cp}} = \beta/2$ and $\beta \leq 1/60$,

$$\frac{1}{1+\beta} + \delta_{\text{cp}} < 1, \qquad \frac{1+\beta/4}{(1+\beta/2)^2} + \delta_{\text{cp}} < 1.$$

Thus the lifted wire value stays strictly in the intended interval even after the allowed price residual. The clipped decoding is therefore inactive on gate outputs:

$$\Pi_{[0,1]}(P_0 - p) = P_0 - p.$$

Proof. The displayed expressions are the normalized price drops over the largest one-writer gate range after the slope match. Both are strictly below one for $\beta > 0$. Average pools use a smaller per-writer slope and the same total range bound. It remains to check that the final residual budget still keeps decoding inside the unclipped band.

$$\frac{1}{1+\beta} + \frac{\beta}{2} < 1$$

is equivalent to $\beta(1-\beta) > 0$, and

$$\frac{1+\beta/4}{(1+\beta/2)^2} + \frac{\beta}{2} < 1$$

is equivalent, after multiplying by $(1+\beta/2)^2$, to

$$\frac{3\beta}{4} + \frac{\beta^2}{2} + \frac{\beta^3}{8} < \beta + \frac{\beta^2}{4},$$

or $2 + \beta < 2/\beta$, which holds for $0 < \beta \leq 1/60$. Hence no additional projection error appears in the final parameter regime.

Lemma 19 (macro error constants). *Let $r = 2\delta + \beta$, where the two δ terms account for liquidation and price residuals and β is the curvature error from Lemma 17. For the lifted constant-product gadgets,*

$$e_\neg \le (2-\beta)r, \qquad e_+ \le (15-8\beta)r \le 30\delta + 15\beta.$$

Proof. A NOT residual is amplified by at most $2-\beta$, giving $e_\neg \le (2-\beta)r$. The SUM macro composes two complements, an average, a complement of the average, and a final gain-2 writer. The errors satisfy

$$e_{\bar{U}} = e_{\bar{W}} = (2-\beta)r, \quad e_A = 2(2-\beta)r, \quad e_{\bar{A}} = 3(2-\beta)r,$$

and the final writer gives

$$e_O = 2e_{\bar{A}} + (3-2\beta)r = (15-8\beta)r.$$

Since $\beta \le 1$, this is at most $30\delta + 15\beta$. Generalized-circuit satisfaction is local, so these constants do not accumulate with the circuit size.

Final constants. Let $\kappa \le 1$ be the rational source accuracy constant after the harmless decrease in Theorem 5. Set

$$P_0 = 60/\kappa, \qquad \beta = \kappa/60, \qquad \delta_{\mathrm{cp}} = \beta/2 = \kappa/120.$$

Every δ_{cp}-clearing decodes to a gate violation at most

$$30\delta_{\mathrm{cp}} + 15\beta = 30 \cdot \frac{\kappa}{120} + 15 \cdot \frac{\kappa}{60} = \frac{\kappa}{2} < \kappa.$$

This proves the robustness claim used in Theorem 5.

F Source Problem and Fan-Out

The restricted source problem in Lemma 3 is taken as a black box. The load-bearing citation is the full version of Budish et al. [8, Appendix D.1, Definitions 10–11 and Theorem 3], which proves constant-error PPAD-completeness for simplified circuits containing only G_+ and $G_\neg$, even with fan-out two. Their formal assignments may take values in $[0, 1+\kappa]$. The clipped $[0,1]$ formulation used here is harmless: after fixing unproduced source nodes in $[0,1]$, the resulting clipped SUM/NOT map has an exact fixed point in $[0,1]$, and every $[0,1]$-valued κ-solution is also a valid solution for their formulation. After decreasing κ by a constant factor, we may also pad the source circuit by constant-length two-NOT copy paths so that every gate output is a fresh wire distinct from its inputs. A source node not produced by any gate is represented in the linear construction by an asset with baseline price $p_v^0 = 2$ and no collateral writer, so $p_v = 2$ and its decoded value is zero. In the constant-product construction, the analogous untraded asset has price P_0. This preprocessing only changes the circuit size by a constant factor and matches the liquidation model's requirement

$a_i \neq b_i$. Filos-Ratsikas et al. [7] and Rubinstein [34] provide related restricted-gate and generalized-circuit hardness background. Our construction does not require bounded fan-out because debt-leg reads are non-invasive.

The unbalanced-source promise used in Lemma 4 is not a target-side graph tag. It is obtained before translation by adding the two gates $a = 1 - s$ and $b = T_{[0,1]}(s + a)$ through an original source wire s. This creates a signed source cycle $s - a - b - s$ with one NOT edge. Lemma 6 then maps that source frustration to an odd collateral-debt cycle in the constructed borrowing graph.

G Subtraction-Free Structure of the Balanced Side

The sign-balanced map in Theorem 1 has the special form

$$y = \Pi_{[0,u]}(Ay + c), \qquad A \geq 0.$$

It is therefore monotone and subtraction-free. It can form nonnegative weighted sums and clip against constants, but it has no direct primitive for comparing two variables. Boolean monotone fragments such as AND/OR have easy least fixed points by finite Kleene iteration, while large-domain Tarski hardness constructions typically use min/max or other variable-comparison operations. Implementing such operations appears to require an anti-monotone dependence, which is precisely what the bipartite sign transform rules out.

LCP location. Writing $B = I - A$, the fixed-point condition is the box complementarity system

$$\begin{aligned} y_i = 0 &\Rightarrow (By + d)_i \geq 0, \\ 0 < y_i < u_i &\Rightarrow (By + d)_i = 0, \\ y_i = u_i &\Rightarrow (By + d)_i \leq 0, \qquad d = -c. \end{aligned}$$

Because $A \geq 0$, B is always a Z-matrix, and the equivalences

$$\begin{aligned} B \text{ is a } P\text{-matrix} &\iff B \text{ is a nonsingular } M\text{-matrix} \\ &\iff \rho(A) < 1 \end{aligned}$$

locate the regimes. Small impact is an M-matrix box LCP, while large impact has a Z-matrix base that need not be a P-matrix. The unbounded Z-matrix LCP is polynomial for every Z-matrix. The obstruction here is the upper bound, whose standard-LCP embedding introduces the positive block shown in Appendix C. The correct classical object is therefore bounded Z-matrix complementarity rather than the ordinary embedded LCP. Proposition 2 invokes the least-element theory for that bounded problem. The small-impact M-matrix regime remains useful because it gives uniqueness and contraction.